DS 3021 HW : General Linear Models

4.

1) $\hat{y} = b \cdot x = b_0 + \sum_{k=1}^{K} b_k x_k$

derivative w/ respect to the k-th feature : $\boxed{\frac{\partial \hat{y}}{\partial x^k} = b_k}$

For a 1 unit change in my feature, the prediction changes by $\boxed{b_k \text{ units}}$. This rate of change was found with calculating the derivative of $\hat{y}$.

2) $\hat{p}_i = \frac{e^{b \cdot x}}{1+e^{b \cdot x}}$ $L = b \cdot x \leftarrow$ from the hint $\frac{\partial}{\partial L} \frac{e^L}{1+e^L} = \frac{e^L}{1+e^L}\left(1 - \frac{e^L}{1+e^L}\right)$

$$\boxed{\frac{\partial \hat{p}}{\partial x_k} = \frac{\partial \hat{p}}{\partial L} \cdot \frac{\partial L}{\partial x_k} = \hat{p}(1-\hat{p}) \cdot b_k}$$

As x changes, my answer also changes. This is because the rate of change is calculated by deriving $\hat{p}$. This operation depends on x (as seen in $L = b \cdot x$).

The derivative of this prediction is different from my answer in part 1. In part 1, my prediction always changes by b_k units. On the other hand, the change for part two is non-linear since b_k is multiplied with $\hat{p}(1-\hat{p})$.

3) $\log\left(\frac{\hat{p}}{1-\hat{p}}\right) = b \cdot x$ $\rightarrow$ $f(x) = \log\left(\frac{\hat{p}}{1-\hat{p}}\right) = b_0 + b_1 x_1 + \dots + b_k x_k + \dots + b_K x_K$

adding a change of 1 to x_k

$f(X_{new}) = b_0 + b_1 x_1 + \dots + b_k(x_k + 1) + \dots + b_K x_K$

$= b_0 + \dots + b_k x_k + b_k + \dots + b_K x_K$

old - new $= (b_0 + \dots + b_k x_k + b_k + \dots) - (b_0 + \dots + b_k x_k + \dots)$

$= b_k$

A one unit change in x_k increases the log odds ratio by $\boxed{b_k}$.